Lattice Vector Quantization of Generalized Gaussian Sources

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Reference

[1] F. Chen, Z. Gao, and J. Villasenor, "Lattice vector quantization of generalized Gaussian sources," *IEEE Transactions on Information Theory*, vol. 43, no. 1, pp. 92–103, Jan. 1997, doi: 10.1109/18.567652.

[2] T. Fischer, "A pyramid vector quantizer," *IEEE Transactions on Information Theory*, vol. 32, no. 4, pp. 568–583, Jul. 1986, doi: 10.1109/TIT.1986.1057198.

- Lattice-Based Quantizer
- Hopefully followed by lattice on cryptography texts

Overview

- Goal: Analyze $R(D)$ performance of $\mathbb{Z}$ lattice quantization for Generalized Gaussian (GG) sources.
- Key idea: Shell concentration in high dimensions.
- Strategy: Compare asymptotic bounds vs low-rate behavior.

Generalized Gaussian Distribution

$$f_X(x) = C_1 e^{-C_2|x|^\nu}, \quad \nu > 0$$

$$C_1 = \frac{\nu\eta(\sigma,\nu)}{2\Gamma(1/\nu)}, \quad C_2 = (\eta(\sigma,\nu))^\nu, \quad \eta(\sigma,\nu) = \frac{1}{\sigma}\left(\frac{\Gamma(3/\nu)}{\Gamma(1/\nu)}\right)^{1/2}$$

- ν: shape parameter
- σ: standard deviation

Shannon Lower Bound for GG Source

$H(X)$ obtained from differential entropy

$$\begin{aligned} h(\nu) &= C_2 \mathbb{E}[|X|^\nu] - \ln C_1 \\ &= \frac{1}{\nu} 2 C_1 C_2^{-1/\nu} \Gamma\left(1 + \frac{1}{\nu}\right) - \ln C_1 \end{aligned}$$

with

$$H(X|\hat{X}) = \frac{1}{2} \log 2\pi e D$$

and we have

$$R_{\mathrm{SLB}}(D, \nu) = \frac{1}{\nu \ln 2} 2 C_1 C_2^{-1/\nu} \Gamma\left(1 + \frac{1}{\nu}\right) - \log_2 C_1 - \frac{1}{2} \log 2\pi e D$$

$$\Rightarrow D_{\mathrm{SLB}}(R, \nu) = \left(\frac{1}{2\pi e} \cdot 2^{\frac{2h(\nu)}{\ln 2}}\right) 2^{-2R}$$

Transformation (1)

- For N i.i.d. GG components: $\mathbf{X} = \{X_i\}_{i=1}^{N}$
- Norm concentration:

$$\frac{1}{N}\sum_{i=1}^{N}|X_i|^{\nu} \to \mathbb{E}[|X|^{\nu}]$$

- ℓ_ν Norm:

$$L = \left(\sum_{i=1}^{N}|X_i|^{\nu}\right)^{\frac{1}{\nu}}$$

- The norm depends on absolute values of the components x_i, so we begin with the distribution of $|X|$ for a 1D generalized Gaussian:

$$f_{|X|}(x) = 2C_1 \exp(-C_2 x^{\nu}), \quad x \geq 0 \tag{8}$$

Transformation (2)

- Make the substitution $Y = |X|^{\nu}$ and apply a change of variables:

$$f_Y(y) = \frac{2C_1}{\nu} y^{1/\nu - 1} \exp(-C_2 y), \quad y \geq 0 \tag{9}$$

Follows a Laplacian distribution.

Pyramid Vector Quantizer Asymptotic Analysis

- For high rate and dimension, vectors will cluster on a "shell" of constant probability.
- Asymptotic distortion can be described using the distortion of quantizing a uniform source.
- Fischer's original PVQ paper analyzed Z-lattice-based quantization for a Laplacian source.
- These arguments extend naturally to generalized Gaussian (GG) sources.
- As dimension increases, most lattice points lie near the boundary, making the shell approximation asymptotically valid.

Asymptotic Distortion Prediction

$$D_{\text{poly}}(R, \nu) = \sigma_u^2 2^{-2R}$$

$$R_{\text{poly}}(D, \nu) = \frac{1}{2} \log_2 \frac{\sigma_u^2}{D}$$

$$\sigma_u^2 = \frac{1}{12} 2^{2h(\nu)/\ln 2}$$

- Uniform vector assumption on shell: $\mathbb{E}[x^2] = \frac{1}{12}$
- GG gives better MSD than uniform under low-rate

Granular and Overload Distortion

- **Granular distortion**: inside the quantization region
- **Overload distortion**: outside the region
- two different approximations for the truncation boundary to obtain the expression for projection distortion
- At high rate: number of zero components $S \to 0$

Further Questions

- How to choose a proper lattice for a given source
- Generalized Gaussian distribution for trainable model parameters
- Distortion of lattice quantization that is not a $\mathbb{Z}$ lattice

Plans

- Main theme: Conway, Zamir, (Micciancio)
- selected papers